



North South University

Project Report of

Programming language I Lab (CSE115.8)

Submitted to

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Pediatric Clinic Management System

Introduction

The Pediatric Clinic Management System provides an efficient platform for administration to manage staff, and for users to easily access their data. Efficient data storage, user authentication and the inclusion of critical features such as edit, deletion and search in a system are among the main objectives of this project. Other objectives of this program include:

- Implement a secure user authentication system
- View records of users, adding staff, book and reschedule appointments, bill payment, etc
- Create an interactive user-friendly interface for ease of use

By providing a digital platform for efficient data handling, this project aims to improve the management of pediatric clinics that still operate by maintaining physical copies and files. The program provides an error-free, easy to maintain and a much faster foundation for clinics.

Background

In many clinics even to this day, data is stored in form of physical files. Physical storage of data such as that is often unstructured and hard to manage. Simple task such as editing or searching is time consuming, additionally, it is more error-prone. This project's main objective is to digitalize this process, and efficiently store and manage large datasets. Moreover, provide an user-friendly interface that can be used by admin, staff and patients.

Libraries Used:

1. `stdio.h`

- `printf()`: Used for formatted output. It is employed to display messages and information to the user on the console.

- `scanf()`: Used for formatted input. It is utilized for reading input from the user during authentication and data entry.
- `fprintf()`: Writes formatted data to a stream, in this case, used to write staff or patient data to the file.
- `fscanf()`: Reads formatted data from a stream, employed to read staff or patient data from the file.
- `fclose()`: Closes the file stream, used to close files after reading or writing.
- `rewind()`: It is used to move the file position indicator to the beginning of a given file stream and clears the end-of-file and error indicators associated with that stream.
- `feof()`: It is used to check if a file stream's end-of-file indicator has been set.
- `getchar()`: It is used to read a single character from the standard input. Here it is used to clear input buffer from the keyboard.
- `fgets()`: It is used to read a line of a text from a specified input stream.

2. `stdlib.h`

- `remove()`: Deletes a file. Used to remove the original staff detail file when updating it with new information
- `rename()`: renames a file. Used to replace the original staff data file with a temporary file after deleting a staff.

3. `string.h`

- `strcmp()`: Compares two strings. Used for comparing entered usernames and passwords during authentication.
- `strstr()`: Finds the first occurrence of a substring. Used to search for a specified car name or model in the file.

4. `stdbool.h`

- `false` or `true`: Used as flags to determine progress of a function.

These library functions provide essential functionality for file handling, user input/output, string manipulation, and basic error handling in the program.

Purpose and Benefits

This Pediatric Clinic Management System solves a real-world problem; managing patient records, medical data, and billing manually is slow and inconvenient. This program organizes everything into four clear portals, each handling a specific responsibility. The admin portal controls system access and overall clinic oversight. The staff management manages, view records, add and edit staff, search and delete information. Then the Staff portal easily logs into all the staff's personal account. The patient portal allows patients to view only their own records and all information relating to it. Moreover, patients can view, search, reschedule and cancel their appointment if they want. The billing portal automates the financial side; pulling fees from multiple sources to generate accurate bills, processing payments by cash or credit card, and maintaining a logged payment history. Rather than flipping through files or doing manual calculations, staff can search, update, and process records in seconds. Each portal is self-contained, meaning the right people only access what is relevant to them. Built entirely in C using core concepts like file handling, functions, and menu driven logic, the system proves that practical, real-world problems can be solved cleanly without complex tools.

System Features

1. Staff Management Portal

Feature	Description
User authentication	Login into admin account using username and password
View all Records	Display a table of all staff currently in the system
Add Staff	Functionality to add new staff to the system
Search Staff	Search for staff using their name
Edit Staff	Functionality to edit staff details
Delete Staff	Remove staff from system using ID numbers

2. Staff Portal

Feature	Description
User authentication	Login into admin account using username and password
View profile	Display details about the staff

3. Patient Portal

Feature	Description
User authentication	Patients can register and log in using their Patient ID and password.
View Profile	Patients can view their own personal and medical details after login.
View Available Doctors	Patients can see the list of available doctors before booking an appointment.
Book Appointment	Patients can book an appointment by selecting a doctor, date, and time.
View Appointment	Patients can view their previously booked appointments.
Search Appointment	Patients can search appointments by typing part of a doctor's name.
Reschedule Appointment	Patients can view their current appointments and change the date or time.
Cancel Appointment	Patients can cancel a booked appointment if needed.
View Prescription	Patients can view prescriptions given by the doctor.

View Bed Info	Patients can check whether a hospital bed has been assigned to them.
View Bill	Patients can view their billing information and payment status.

4. Billing and Payment Portal

Feature	Description
User authentication	Login into admin account using username and password
Generate Bill	Retrieves patient data from records and calculates the total bill by combining consultation fee, bed charge, and medicine cost
View Bill	Searches for a patient by ID and displays their full itemized bill along with current payment status
Process Payment	Allows staff to select a payment method and marks the patient's bill as paid, updating the records accordingly
View Payment History	Displays a list of all completed payments including patient ID, total amount paid, and payment method used

User Stories

As an administrator, I want to easily manage staff records of the clinic so that information stays accurate and easy to find. I want my portal to be secure so I have my own username and password. After logging in, I have the choice to view all staff details, add staff, search staff by name, edit staff using their ID and delete staff if they are no longer present in the clinic.

As a staff of the clinic, I want to log in and view my own details so that I can confirm my own details in the system. After successfully logging in I can view my ID, name, specialization, phone and email.

As a patient, I want to log in or register into my portal so that I can view and edit my own details. After successful login or registration, I have the options to view my profile, view available doctors, book an appointment, view my already booked appointments, reschedule appointment if there is a need and cancel the appointment as well. Moreover, I can search my appointments, view prescription, view bed information and view my total bill.

As a staff, I want to be able to generate bill for patients and as a patient I want to view my bill details, view payment history and pay my bill using cash or online. After Successful login into billing portal, I can choose any of these options by entering my choice.

Advantages

The Pediatric Clinic Management System offers a much more efficient alternative to managing manual records and files. It is more structured and less prone to error. It has a user-friendly interface that can be used by anyone. Additionally, it speeds up daily processes such as searching, viewing and updating records. Its' strengths lie in the diverse functionality it offers, such as bill payment, appointment booking, editing and viewing details. Moreover, every portal has a required user verification making it a safe and reliable platform to store data. On the contrary, many clinics still use manual records which is often unstructured and error prone.

Basic task such as verifying appointment takes more time than acceptable. Thus, this program can save both time and money for both the clinic and patient

Limitations

- Security features are weak. In portals such as staff management portal and bill and payment portal, the username and password are hardcoded into the code making it unsafe.
- When logging into a portal, there is unlimited chances to enter correct username and password. Standard practice is to offer the user three chances before shutting them out of the portal.
- To find ID number of staff, line number is counted. However, this is susceptible to error in case line numbers don't match.
- Linear search is used to search the database. In case of large datasets, this would be time consuming.
- The search function is case sensitive which makes it less versatile and difficult to use.
- There is no flag to ignore string input when entering ID. This leads to the program breaking down by entering an infinite loop.
- The Patient Portal checks whether a doctor is marked as Available, but it does not validate whether the patient's entered appointment time is actually within the doctor's available time slot. For example, even if a doctor is available from 10AM-2PM, the system may still accept an appointment at 5PM. This is a limitation of the current appointment booking system.

Topics Covered

The C programming language is used to develop the system. It is based on a modular architecture, consisting of individual functions for each feature, which contributes to code organization and reusability. For long term data storage, file handling is used. The project touches many core C concepts such as:

- Control flow (if, switch, loops) to handle menus and logic.
- Functions for modular design (separate functions for add, edit, delete, search, etc).
- Structures to group and maintain related data.
- Array to store multiple records and for reading file data into code to be used efficiently.
- Pointers for passing structures and arrays efficiently.
- File handling to store and load records from a text or csv file.
- Library functions such as string.h to compare, copy, search, etc.
- Input validation and buffer handling to avoid common console input bugs.

Screenshots

```
=== pediatrics clinic hospital management system ===  
  
1. Staff Management Portal  
2. Staff Portal  
3. Patient Portal  
4. Billing & Payment Portal  
5. Exit  
Enter choice: 1  
  
=== Staff Management Portal ===  
Username: admin  
Password: 1234  
  
Admin logged in successfully.  
  
== Staff Management Menu ==  
1. View all records  
2. Add Staff  
3. Search Staff  
4. Edit Staff  
5. Delete Staff  
6. Exit  
Enter choice: |
```

```
"C:\Users\mahir\OneDrive\De X + v
Enter choice: 1
File opened successfully
=====
ID      Name                Age  Staff Type                Availability  Working Hours
=====
001 Dr. Sarah Rahman      38   Pediatrician              Mon-Fri     09:00-17:00
002 Dr. Ayan Khan        42   Pediatrician              Mon-Sat     10:00-18:00
003 Dr. Priya Das        35   Cardiology Specialist     Tue-Thu-Sat 09:00-15:00
004 Nurse Anika Islam   29   Nurse                    Mon-Fri     08:00-16:00
005 Nurse Rafi Ahmed    31   Nurse                    Mon-Fri     16:00-00:00
006 Nurse Tania Roy     27   Nurse                    Mon-Sat     09:00-17:00
007 Dr. David Kim       40   Neurology Specialist      Mon-Wed-Fri 10:00-16:00
008 Rachel Green       33   Physiotherapist          Mon-Fri     09:00-17:00
009 Laura Bennett      28   Psychologist             Tue-Thu-Fri 10:00-16:00
010 Kevin Thomas       30   Medical Assistant        Mon-Sat     08:00-16:00
011 Linda Carter       26   Receptionist             Mon-Sat     08:00-16:00
012 Arjun Mehta        29   Lab Technician           Mon-Sat     08:00-16:00
013 Sophia Martinez    32   Dietitian                Mon-Wed-Fri 10:00-16:00
014 Adam Brooks        34   Radiology Technician      Mon-Sat     09:00-17:00
015 Olivia White       25   Cleaning Staff           Mon-Sat     07:00-15:00
=====
== Staff Management Menu ==
1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit
Enter choice: |
```

```
== Staff Management Menu ==
1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit
Enter choice: 2

Enter Name: Dr.Mahir Daiyan
Enter age: 67
Enter staff type: Plague Doctor
Enter Availability details: NONE
Enter working hours: 25:00-26:00
Successfully added!

== Staff Management Menu ==
1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit
Enter choice: |
```

```
== Staff Management Menu ==
```

1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit

Enter choice: 3

Enter staff name: Dr. Sarah Rahman

Staff found

ID	Name	Age	Staff Type	Availability	Working Hours
001	Dr. Sarah Rahman	38	Pediatrician	Mon-Fri	09:00-17:00

```
== Staff Management Menu ==
```

1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit

Enter choice: |

```
*C:\Users\mahir\OneDrive\De X + v
== Staff Management Menu ==
1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit
Enter choice: 4

Edit:
1. Name
2. Age
3. Staff Type
4. Availability
5. Working Hours
6. Exit
Enter choice: 2

Enter ID: 001
Current age: 38
Enter new age: 40

== Staff Management Menu ==
1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit
Enter choice: |
```

```
== Staff Management Menu ==
```

1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit

Enter choice: 5

Enter staff ID: 015

Staff deleted successfully

```
== Staff Management Menu ==
```

1. View all records
2. Add Staff
3. Search Staff
4. Edit Staff
5. Delete Staff
6. Exit

Enter choice: |

```
"C:\Users\mahir\OneDrive\De  X + v
=== pediatrics clinic hospital management system ===
1. Staff Management Portal
2. Staff Portal
3. Patient Portal
4. Billing & Payment Portal
5. Exit
Enter choice: 2

=== Staff Portal ===
1. Login
2. Back
Enter Choice: 1

ID: 001
Password: S

Successful login.
```

```
=====
ID:          001
Name:        Dr. Sarah Rahman
Specialization: Pediatrician
Phone:       +8801711000001
Email:       sarah.rahman@clinic.com
=====

To logout press 0: 0

=== Staff Portal ===
1. Login
2. Back
Enter Choice: |
```

```
=== Patient Portal ===
1. Login
2. Register
3. Back
Enter choice: 1

=== Patient Login ===
Enter Patient ID: 001
Enter Password: 1234
Login successful.

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |
```

```
=== Patient Profile ===
ID      Name      Height  Weight  Condition  Blood Group  Doctor Assigned
001     Ayaan Rahman  120 cm  25 kg   Fever      A+           Dr. Sarah Rahman

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |
```

```
=== Available Doctors ===
ID      Name      Specialization  Time
-----
201     Dr. Sarah Rahman  Pediatrics      10AM-2PM
202     Dr. Ayan Khan     General Medicine 2PM-6PM
203     Dr. Priya Das     Cardiology      9AM-1PM

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |
```

```
=== Book Appointment ===

=== Available Doctors ===
ID      Name      Specialization  Time
-----
201     Dr. Sarah Rahman  Pediatrics      10AM-2PM
202     Dr. Ayan Khan     General Medicine 2PM-6PM
203     Dr. Priya Das     Cardiology      9AM-1PM
Enter Doctor ID: 201
Enter Date (DD-MM-YYYY): 02-01-2000
Enter Time (HH:MM): 13:00
Appointment booked successfully.

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |
```

```
=== My Appointments ===
PatientID  DoctorID  Date        Time        Status
001        201         26-04-2026  10:00       Booked
001 201 02-01-2000 13:00 Booked
```

```
=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |
```

```
=== Search Appointment By Doctor Name ===
```

```
=== Available Doctors ===
ID      Name                Specialization      Time
-----
201     Dr. Sarah Rahman       Pediatrics          10AM-2PM
202     Dr. Ayan Khan           General Medicine    2PM-6PM
203     Dr. Priya Das           Cardiology          9AM-1PM
```

```
Enter part of doctor name: 201
Doctor not found or doctor is not available.
```

```
=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |
```

```
=== Your Current Booked Appointments ===
Doctor ID: 201 | Date: 26-04-2026 | Time: 10:00 | Status: Booked
Doctor ID: 201 | Date: 02-01-2000 | Time: 13:00 | Status: Booked
```

```
=== Reschedule Appointment ===
Enter Doctor ID from the list above: 201
Enter Current Date from the list above: 26-04-2026
```

```
=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: 7
```

```
=== Your Current Booked Appointments ===
Doctor ID: 201 | Date: 26-04-2026 | Time: 10:00 | Status: Booked
Doctor ID: 201 | Date: 02-01-2000 | Time: 13:00 | Status: Booked
```

```
=== Cancel Appointment ===
Enter Doctor ID from the list above: 201
Enter Date from the list above: 02-01-2000
```

```

=== My Bed Info ===
Patient ID      Bed No      Ward      RoomNo      Admission Date      Discharge Date      B
ed Charge      Status
No bed assigned.

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |

```

```

=== My Prescription ===
PatientID      DoctorID      Diagnosis      Medicine      Dosage      Advice
001            201           Fever          Paracetamol   TwiceDaily   Rest_And_Drink_Water

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |

```

```

=== My Bill ===
PatientID      Consultation Fee      Bed Charge      Medicine Cost      Total Bill      Status
001            500                   0               300                800             Paid

=== Patient Menu ===
1. View Profile
2. View Available Doctors
3. Book Appointment
4. View Appointment
5. Search Appointment
6. Reschedule Appointment
7. Cancel Appointment
8. View Prescription
9. View Bed Info
10. View Bill
11. Logout
Enter choice: |

```

```
"C:\Users\mahir\OneDrive\De  X + v - □ X

=== pediatrics clinic hospital management system ===

1. Staff Management Portal
2. Staff Portal
3. Patient Portal
4. Billing & Payment Portal
5. Exit
Enter choice: 4

=== Billing & Payment Portal Login ===
Username: billing
Password: 1111

Login Successful.

=== Billing & Payment menu ===
1. Generate Bill
2. View Bill Details
3. Process Payment
4. View Payment History
5. Exit
Enter choice: |
```

```
=== Generate Bill ===
Enter Patient ID: 3

=====
                        GENERATED BILL - PATIENT 003
=====
Consultation Fee   : 500 BDT
Ward               : Cardiology
Bed No            : B201
Room No           : 21
Admission Date    : 25-04-2026
Discharge Date    : NotDischarged
Bed Charge        : 1800 BDT
Diagnosis         : Heart_Murmur
Medicine          : Atenolol
Dosage            : OnceDaily
Medicine Cost     : 700 BDT
-----
TOTAL BILL        : 3000 BDT
Status           : Paid
=====

=== Billing & Payment menu ===
1. Generate Bill
2. View Bill Details
3. Process Payment
4. View Payment History
5. Exit
Enter choice: |
```

```
=== View Bill Details ===
Enter Patient ID: 3

=====
                        BILL DETAILS - PATIENT 003
=====
Consultation Fee   : 500 BDT
Bed Charge        : 1800 BDT
Medicine Cost     : 700 BDT
-----
Total Bill        : 3000 BDT
Payment Status    : Paid
=====

=== Billing & Payment menu ===
1. Generate Bill
2. View Bill Details
3. Process Payment
4. View Payment History
5. Exit
Enter choice: |
```



```

=== Process Payment ===

Select Payment Method:
1. Cash
2. Credit Card
Enter choice: 1
Payment method: Cash

Enter Patient ID to be marked as Paid: 7

=====
                PAYMENT RECEIPT
=====
Patient ID      : 007
Total Bill      : 3300 BDT
Payment Method  : Cash
Status          : Paid
=====
Payment status updated successfully.

=== Billing & Payment menu ===
1. Generate Bill
2. View Bill Details
3. Process Payment
4. View Payment History
5. Exit
Enter choice: |

```

```

=== Payment History (Paid Bills) ===

=====
PatientID  Consult  BedCharge  Medicine  Total  Status
=====
001        500      0          300       800    Paid
002        500      0          250       750    Paid
003        500     1800       700      3000    Paid
004        500      0          250       750    Paid
005        500      0          400       900    Paid
006        500      0          350       850    Paid
007        500     2000       800      3300    Paid
010        500      0          300       800    Paid
012        500      0          500      1000    Paid
014        500     1500       650      2650    Paid
=====
Total paid records: 10

=== Billing & Payment menu ===
1. Generate Bill
2. View Bill Details
3. Process Payment
4. View Payment History
5. Exit
Enter choice: |

```

Future Work

Security features need to be enhanced. Instead of storing login credentials in plain text, they can be encrypted. Moreover, a sorting algorithm such as bubble sort can be used to order the data, enabling better searching algorithm such as binary search to be used. This will result in the reduction of time it takes the program to process. Input buffer bugs need to be addressed by setting up flags. Furthermore, the console-based UI can be replaced with a graphical or web-based interface to make the system more user friendly.

Team Member Contributions

Name	Role
Mahir Daiyan Hisham	Responsible for developing the Staff Portal. Features: Staff Login System that allows staff members to authenticate using their Staff ID and password. The system reads credentials from staff_login.txt View Staff Profile after a successful login, the profile display feature that retrieves and presents the logged-in staff member's details from staff_profile.txt
Sadman Sakib	Responsible for implementation of the staff management portal. The portal consists of 6 features which are: view record, add staff, search staff, edit staff and delete staff. Used file handling with records.csv and records.txt. Developed a function to find maximum ID and another function to transfer csv file details into an array for easier access of file data in code. Primarily responsible for preparing the project report and its' outline.
Shormila Sawom	Developed and implemented the full Patient Portal module independently. Implemented patient registration, login, profile viewing, available doctor viewing, appointment booking, appointment viewing, appointment

	search, rescheduling, cancellation, prescription viewing, bed information viewing, and bill viewing. Used file handling with patient_login_file.txt, patientfile.txt, doctors_file.csv, appointments_file.txt, prescriptions_file.txt, beds_file.txt, and bills_file.txt. Added doctor availability checking, partial doctor name search, and previous appointment display before rescheduling or cancellation. Assisted in writing the project report.
Zahra Mahbuba Rahman	Co-proposed the idea of a Pediatric Clinic Management System and helped shape the overall concept during the brainstorming phase. Developed and implemented the Billing and Payment Portal module, including bill generation, bill viewing, payment processing with cash and credit card options, and payment history tracking. Handled file reading and updating across bills_file.txt, beds_file.txt, and prescriptions_file.txt to retrieve and calculate patient bills accurately. Assisted in writing and structuring the project report.